

CATASTROPHIC

1. (i) A.
 (ii) B.
 (iii) False.
 (iv) True.
 (v) True.

2. (A), (i) System $\div$ is an arrangement in which all its units assembled work together according to set of rule eg. watch.

(ii) Embedded $\div$ is the combination of hardware and software ~~for~~ which is designed to perform a particular task at a given time.

(iii) Model-driven development (MDD)
 - is a format to write and implement software quickly, effectively and at minimum cost.

(iv) Abstract System $\div$ is a theoretical model of a system that focuses on its essential characteristics and behavior without considering its implementation details.

(B) Discuss the on rate Monotonic scheduling.

Rate Monotonic Scheduling $\div$ is a scheduling algorithm used in real-time system to prioritize tasks based on their periods. Tasks with shorter periods are given higher priority than tasks with longer periods.

(C) Three key approaches to performance improve most for general purpose (GP) computing system:

Ans:

- (i) Increasing clock speed
- (ii) Adding more cores
- (iii) Improving memory bandwidth

Why GP systems are unsuitable for real-time system:

- (i) Unpredictable performance
- (ii) Interrupt handling
- (iii) Scheduling algorithms

Diagram of sensor and actuator:

Description:

Sensor process; Continuously monitors the physical environment and converts the measured data into digital signals that can be processed by the computer.

Actuator process; Receives control signals from the computer and converts them into physical actions such as controlling a motor.

Major difference between sensor and actuator:
Sensor; A sensor measures physical quantities and converts them into electrical signals.

Actuator; An actuator converts electrical signals into physical actions.

Why we need System analysis and design:

- (i) To understand the problem
- (ii) To design a solution
- (iii) To ensure quality
- (iv)

Difference b/w system analysis and system design.
System analysis; Focuses on understanding the problem, gathering the requirements and defining the system's scope.

System design; Focuses on creating a solution, including the architecture, components and interfaces.

Does highly secure software imply highly safe software?

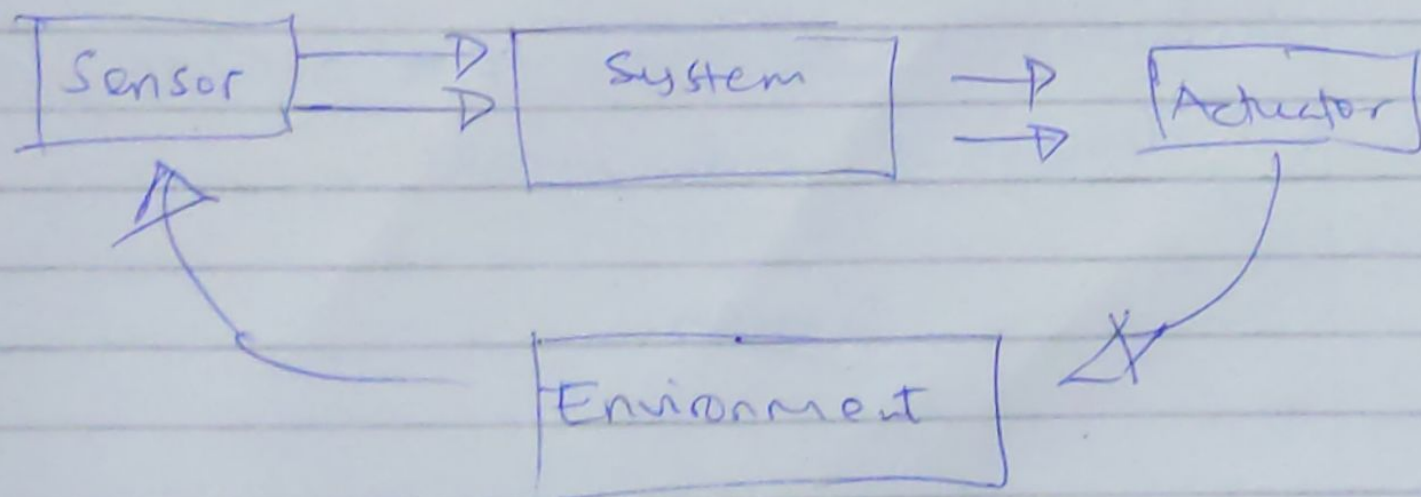
Ans:

NO, security focuses on protecting data and systems from unauthorized access and malicious attacks.

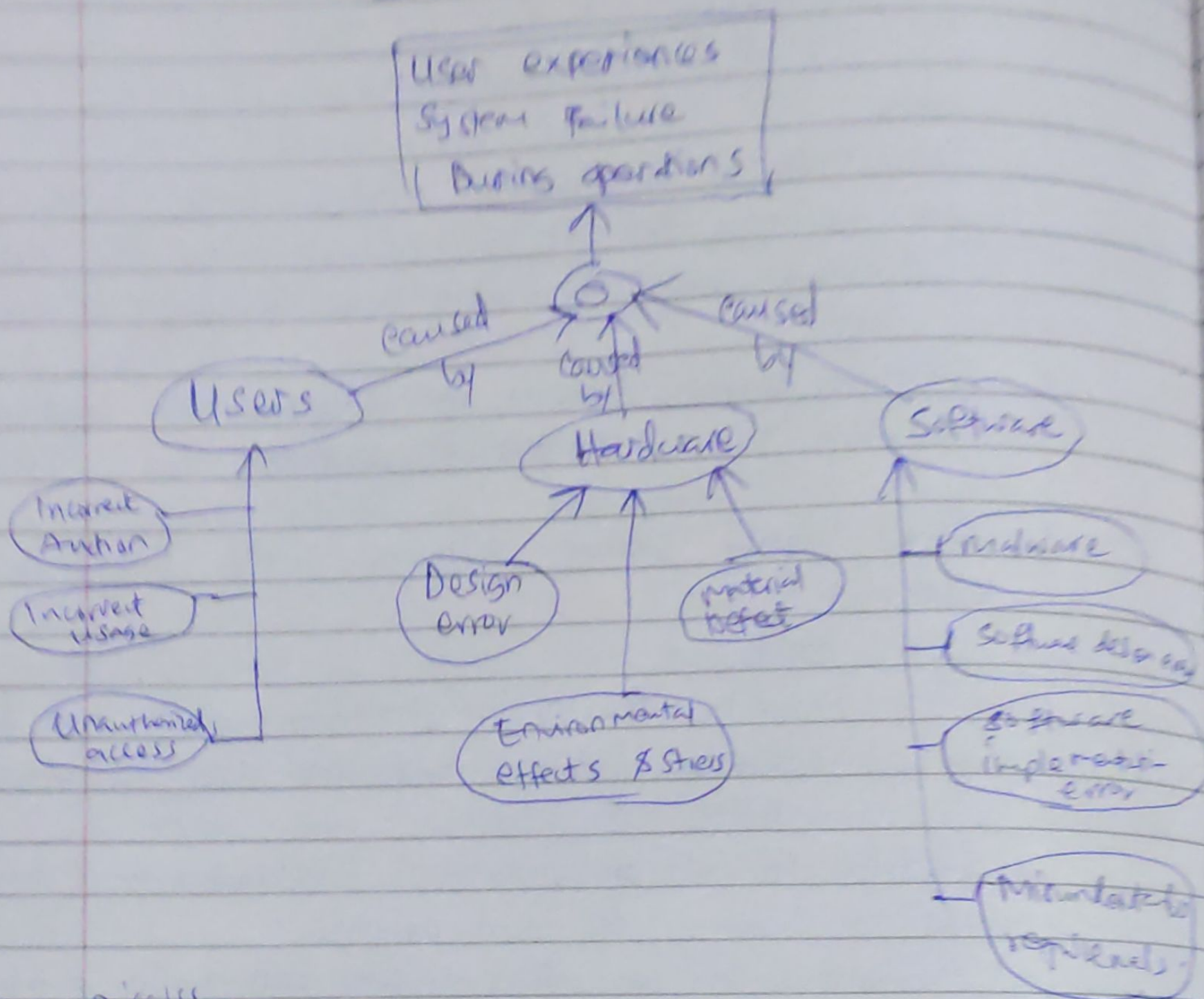
While

Safety focuses on preventing accidents and minimizing risks to human life and property.

A system can be secure but still have safety issues if it fails to handle unexpected situations or malfunctions properly.



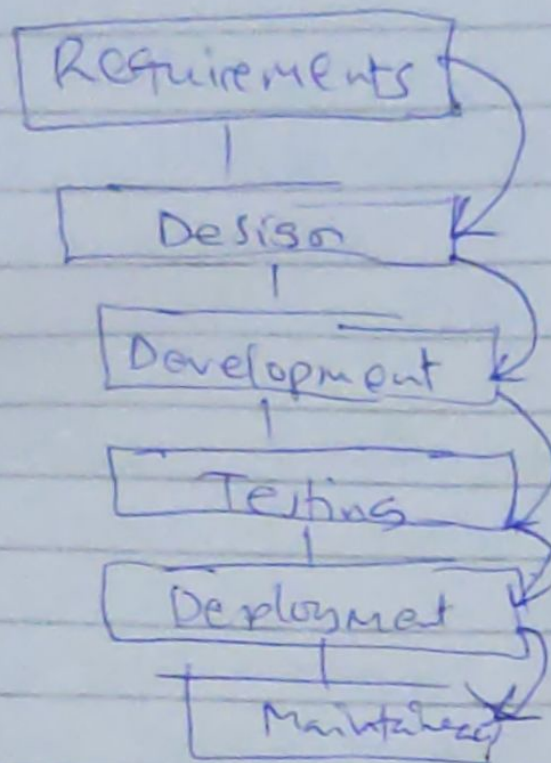
General Causes of System Failure;
Ans:



Discuss
Safety in software development methodologies.
Ans:

- (i) perform safety analysis (hazard analysis, provide hazard report)
- (ii) Implement the results of the analyses.
- (iii) Track and verify changes.

Diagram:



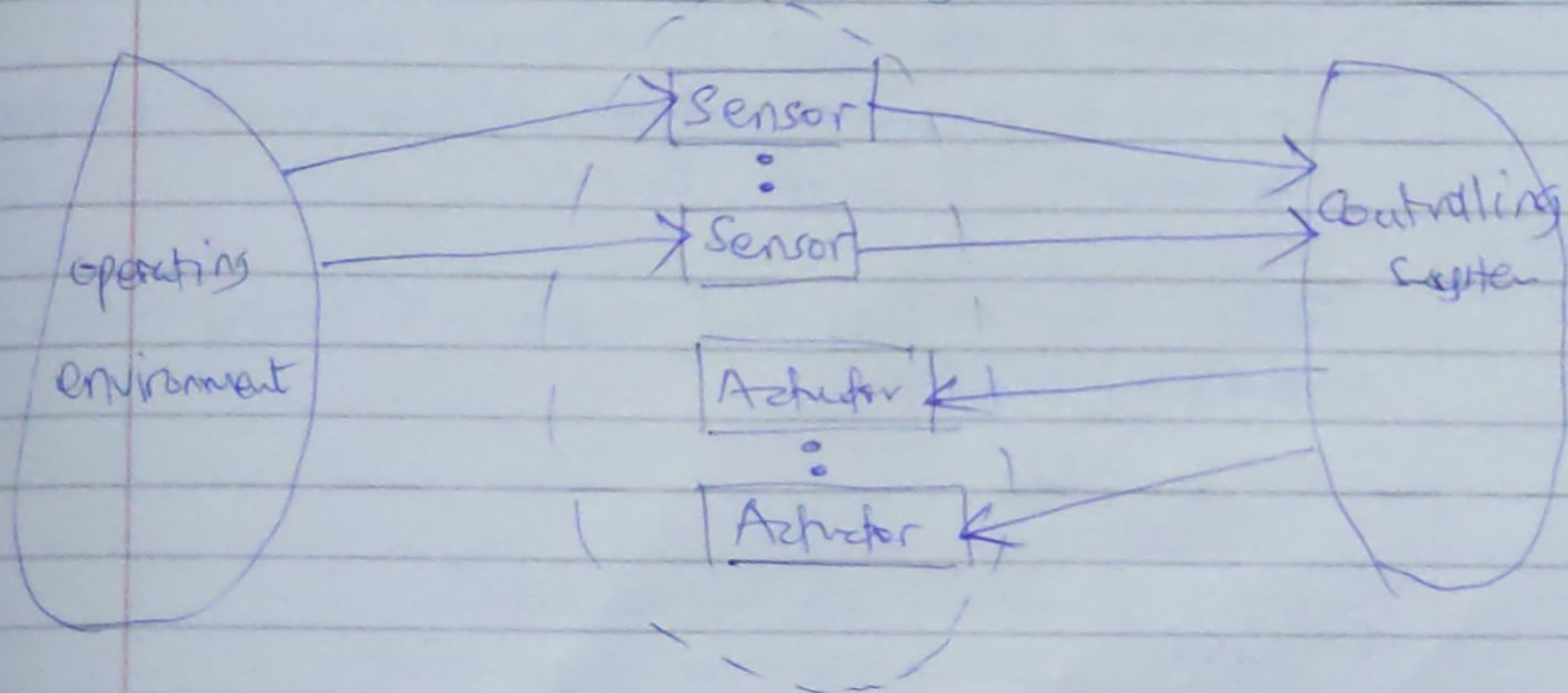
List three (3) Risk consequences of software safety.
ans:

- (i) loss of life or injuries
- (ii) Property damage
- (iii) Economic loss.

With diagram and example, explain the simplified typical real-time system.

Ans:

Controlled System



Example: Car

Mission: Reaching the destination safely

Controlled system: Car

Operating environment: Road conditions and other cars

Controlling system:

- Human driver: sensors - Eyes and ears of the driver
- Computer: sensors - Cameras, infrared receiver, laser rangefinder.

Controls: Accelerator, steering wheel, brake-pedal

Actuators: Wheels, engines and brakes.

List 4 rules for creating safer software:

ans:

i) Communicate

ii) Follow good software engineering practices & procedures.

iii) Perform safety and development analysis

iv) Incorporate appropriate software development Methodologies techniques & design features.

State the 4 pros and cons of Model-Driven Architecture (MDA).

Soln:

Cons:

i) PIM models can't be validated

ii) PSM are Architecture dependent

iii) Generated code still require further development.

iv) Transformation/code generation is an integral part of its development process.

PROS:

- i) PIM models are Reusable
 - ii) Marginal reduction in development time
 - iii) ^{use} Methodical approach to software development
- (iv) It has more direct input from Application analysts.

Types of serial sensor.

- i) RS-232
- ii) I2C

Two types of actuator;

- i) Electric actuators
- ii) Pneumatic actuators.

Four benefits of Machine Vision systems;

- (i) Increased accuracy
- (ii) Faster throughput.
- (iii) Improved quality control.
- (iv) Reduced labor costs.